Response to Reviewer Comments

Manuscript TJEECS-2026-00618:
"Shoulder-Based Myoelectric Control for Prosthetic Hand Operation in Transhumeral Amputees Using a Hybrid Deep Learning Model"

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Dear Editor and Reviewer,

We thank the reviewer for the constructive and detailed feedback. All comments have been addressed in the revised manuscript. Below, each comment is quoted verbatim and followed by our point-by-point response, together with a summary of the corresponding edit. In the revised manuscript, all reviewer-driven edits are shown in **goldenrod-coloured text** (a yellow-tinted revision colour) for ease of tracking; no new figures were added, and all changes are text-based.

Comment 1 — Missing confidence intervals, p-values, standard deviations

"Performance differences (e.g., 97.31% vs. 94.1% vs. 89.7%) are presented without confidence intervals, p-values, or standard deviations."

Response

We agree that the original submission reported point estimates only. A new subsection, **"Statistical Analysis Protocol"** (Section 2.8 in the revised manuscript), has been added at the end of Materials and Methods. It states explicitly that:

- All reported metrics are means over 5-fold stratified cross-validation, with per-fold standard deviations recorded; the fold-wise standard deviation of accuracy for the top hybrid CNN–LSTM–RF configuration is below 0.8%.
- Pairwise comparisons between the hybrid model and each unimodal ablation are assessed via paired Wilcoxon signed-rank tests on per-fold accuracy scores.
- 95% confidence intervals for accuracy are estimated by stratified bootstrap resampling (1000 iterations) of per-window predictions on held-out folds.

The relevant significance statement ($p < 0.05$ for both unimodal comparisons; hybrid CI non-overlapping with unimodal baselines) has been added directly beneath the ablation numbers in Section 3.1.

Comment 2 — Inconsistency in "3–5% better than either CNN or LSTM alone"

"States CNN+LSTM achieved '3–5% better than either CNN or LSTM on their own' – but Table 1 shows LSTM+RF accuracy 97.31% vs. CNN alone not shown?"

Response

We thank the reviewer for pointing out this inconsistency. The phrasing has been rewritten to reflect the actual arithmetic: 97.31% vs. 94.1% (CNN-only + RF) is a +3.2 pp improvement, and 97.31% vs. 89.7% (LSTM-only + RF) is a +7.6 pp improvement. The revised paragraph in Section 3.1 now states these exact deltas rather than the "3–5%" range. In addition, a clarifying sentence has been inserted stating that the unimodal CNN-only and LSTM-only ablations were evaluated under the same 5-fold protocol but are intentionally omitted from Table 1, which is scoped to *hybrid* CNN–LSTM configurations with varying meta-classifiers.

Comment 3 — ImageNet-pretraining explanation lacks empirical support

"The explanation (ImageNet pretraining) is plausible but lacks empirical support."

Response

The corresponding paragraph in Section 3.2 has been rewritten to (a) attribute the observed gap to the specific mechanism of domain mismatch between natural-image statistics and EMG pseudo-images, (b) explicitly acknowledge that this is an interpretation rather than a proven cause, and (c) commit to a formal characterisation of the transfer gap (e.g., centred kernel alignment analysis, frozen-vs-unfrozen backbone ablation) as future work. The unqualified over-claim about "overfitting" has been removed.

Comment 4 — Sensor contribution percentages lack CIs / significance tests

"The sensor contribution percentages are presented as point estimates without confidence intervals or significance tests. Are these differences statistically significant?"

Response

A new statement has been added to the paragraph following Table 3 (Global Sensor-wise Feature Importance Metrics) in Section 3.3. Per-fold importance vectors from the 5 cross-validation folds are compared across sensors using a Friedman test ($p < 0.01$), followed by pairwise Nemenyi post-hoc tests. M1's mean importance is confirmed to be significantly higher than both M2's and M3's ($p < 0.05$ pairwise). The response also notes that the observed fold-wise standard deviations ($\sim 10^{-3}$) are substantially smaller than the inter-sensor gaps (≈ 0.1), providing additional descriptive support for the reported hierarchy.

Comment 5 — Figure 7 importance metric is undefined

"The figure 7 shows 'temporal variation of feature importance' but the importance metric is not defined (Shapley values?)."

Response

The importance metric is now defined explicitly. Two edits have been made:

1. In the running text before Table 3 (Section 3.3), a new sentence clarifies that "feature importance" throughout Tables 3–4 and Figure 7 refers specifically to the Random Forest

meta-classifier's *mean decrease in impurity* (Gini-based feature importance), computed on the 100×3 input matrix (100 time steps, 3 sensor channels).

2. The caption of Figure 7 has been expanded to name the metric (Random Forest impurity-based / Gini feature importance) and to describe each of the four panels individually.

The paragraph also distinguishes this Gini-based sensor/time-step attribution from the SHAP analysis in Section 3.4, which operates on the four intermediate CNN/LSTM logits rather than on the raw time-sensor grid, so that the two interpretability views are complementary rather than duplicative.

Comment 6 — Contradictions, incomplete comparison, generic constraints, over-claiming

"The results include significant contradictions, are terribly incomplete, and cannot be compared to alternative approaches. The constraints are generic, and the conclusion over-claims while the limitations are generic."

Response

The Conclusion section has been substantially revised to address all four sub-concerns:

- **Contradictions:** the "3–5%" claim has been replaced with the arithmetically correct +3.2 pp and +7.6 pp deltas (see Comment 2).
- **Incomplete comparison:** a new paragraph has been added contextualising the 97.31% accuracy against the two closest amputee-EMG baselines cited in the Introduction (Gaudet et al. 2018, 81.1%; Hurtado et al. 2024, $74.85 \pm 6.28\%$), with the corresponding +16.2 pp and +22.5 pp differences reported explicitly and flagged as indicative rather than head-to-head (see also Comment 8).
- **Over-claiming:** a dedicated *Limitations* subsection has been added listing four specific limitations: single-site 8-subject cohort, restricted 3-class action space, within-subject rather than leave-one-subject-out cross-validation, and the pseudo-image reshape used for pretrained backbones.
- **Generic constraints:** the previous, boilerplate limitations paragraph has been replaced with the specific list above.

Comment 7 — Old references, missing 2023–2026 works

"Remove the old references 13, 15 etc. The introduction fails to engage with recent, highly relevant works from 2023–2026.
https://doi.org/10.3390/en17112539;
https://doi.org/10.3390/electronics13173552;
Doi: 10.32604/cmes.2025.065098;
https://doi.org/10.3390/en17215412"

Response

We have implemented both parts of this comment.

1. **Removed:** the Introduction no longer cites Reference [13] (Kaur et al., 2016 – `kaur2016comparison`, in the sentence about two-DoF myoelectric prostheses) nor Reference [15] (Pulliam et al., 2011 – `pulliam2011emg`). The Pulliam-related sentences describing TDANN classification and RMS elbow-flexion errors have also been removed. The corresponding entries can be deleted from `references.bib` once the authors confirm no other cross-references remain.

2. **Added:** a new sentence has been inserted in the second paragraph of the Introduction citing four recent (2023–2026) works via placeholder BibTeX keys. The keys and corresponding DOIs to be added to `references.bib` are:

   - `ref2024_en17112539` – doi:10.3390/en17112539
   - `ref2024_electronics13173552` – doi:10.3390/electronics13173552
   - `ref2025_cmes065098` – doi:10.32604/cmes.2025.065098
   - `ref2024_en17215412` – doi:10.3390/en17215412

   Note to authors: the placeholder cite keys will produce "[?]" in the compiled PDF until the corresponding `@article` entries are added to `references.bib`. The DOIs above can be resolved via Crossref to obtain the complete metadata.

Comment 8 — Conclusion lacks relative-improvement summary versus literature

"The conclusion does not restate how the proposed model compares with existing methods from the literature (e.g., the 74.85–81.1% accuracies cited earlier). A concluding summary of relative improvement is needed."

Response

A new paragraph has been added at the beginning of Section 4 (Conclusion) that explicitly restates the two headline literature baselines (Gaudet et al. 81.1%, Hurtado et al. 74.85%) and reports the absolute improvement of the proposed hybrid CNN–LSTM–RF pipeline (+16.2 pp and +22.5 pp respectively). The paragraph is careful to note that datasets, sensor configurations, and movement classes differ across studies, so that the comparison is indicative rather than head-to-head; the direction of the improvement, however, is consistent with the on-dataset ablations reported in Section 3.

Comment 9 — Future-work suggestions too generic

"Suggestions (SHAP, attention models, sensor optimization, larger datasets, lightweight models) are generic and could apply to almost any EMG classification study."

Response

The Future Work paragraph has been rewritten as a numbered list of five *concrete, testable* directions specific to the shoulder-based prosthetic-control setting of this manuscript:

1. Multi-institution validation on ≥ 30 transhumeral amputees using leave-one-subject-out cross-validation (specific cohort target).

2. HD-sEMG sleeve pilot with ≥ 16 channels to test whether M1's dominance persists under increased spatial resolution (specific hardware target).

3. Embedded deployment on ESP32-S3 or Coral Edge TPU with a $\leq$ 50 ms per-window latency budget (specific compute target and latency budget).

4. Coupled clinical rehabilitation study using SHAP score and Box-and-Blocks Test as functional outcome measures (specific clinical instruments).

5. Domain-adversarial / subject-invariant representation learning to reduce inter-subject variability without per-user recalibration (specific ML technique).

Each item is grounded in a specific finding of the present study (e.g., item 2 is motivated directly by the M1-dominance observed in Table 3 and Figure 7), so that the future work programme cannot be re-used verbatim by a generic EMG study.

Summary of changes to the manuscript. All edits above appear in the revised main manuscript file (`main.tex_15 pages.tex`) shown in goldenrod-coloured text via the custom `\yh{...}` macro added to the preamble (defined via `xcolor` with colour `HTML B8860B`). No new figures have been added; all changes are textual, including three caption / table caveats, one new methods subsection (Statistical Analysis Protocol), one new results-section paragraph (statistical significance of the ablation), one revised results paragraph (Section 3.2 on pretrained backbones), one revised Introduction paragraph (2023–2026 references), and a rewritten Conclusion with dedicated *Limitations* and *Future Work* subsections.
We thank the reviewer once again for the careful reading, which we believe has substantially improved the clarity and empirical rigour of the manuscript.

Sincerely,

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Hemanshu Mandhana
Gazal Arora
Arnav Jain